

DATA PROCESSING & ETL ASSIGNMENT 01

Question 1: Describe different types of data sources used in ETL with suitable examples.

Ans: ETL Stands for Extract Transform and Load.

It helps to load data from different different sources or locations. And tries to transform data into

Correct form.

Types of sources used in ETL

XML ,JSON,CRM ,Web applications , Cloud storage and etc.

Question 2: What is data extraction? Explain its role in the ETL pipeline.

Ans2: Data Extraction means taking out data from different different sources or collecting data

From different sources.

ETL pipeline is the process of taking data from one source to another destination.

1.Extract: It means taking out the information from the specific source Excel files, APIs, CSV or cloud applications etc.

2.Transform: It means transforming the data by making it clean and converting into useful formats, Handling missing values, filtering, sorting etc

3.Load: It means loading to the targeted system such as warehouses or databases where it can

Be used for reporting or analysing.

Question 3: Explain the difference between CSV and Excel in terms of extraction and ETL usage.

Ans3:CSV is a simple text- based file containing rows and column. Data extraction is easy and fast.

Commonly used in ETL pipeline source file. Lightweight and easy to automate.

Excel is a spreadsheet file that can contain multiple sheet, formulas, charts, formatting etc.

Data extraction is more difficult or more complex while extracting data from multiple sheets.

More feature-rich comparatively heavier.

Question 4: Explain the steps involved in extracting data from a relational database.

Ans4:Steps involved in extracting data

1. Connecting to databases: First we connect the relational databases through connection tools.
2. Identify the data : which table rows columns are needed for data.
3. Write a query : writing the query for specific data requirement .
4. Filter data : extract the data only which is required.
5. Extract the data : after extracting the data from databases then transferred to ETL process.
6. Stored the extracted data : into the warehouses or cloud databases.

Question 5: Explain three common challenges faced during data extraction.

Ans5: 3 challenges faced during data extraction

1. Large data volume: it means data with large information takes too much time for loading and extracting and consume more storage.
2. Data quality issues: data may contain missing values, null values, duplicate values and incorrect Formats which can create issues in data extractions.
3. Different data sources: data may come from different data sources like excel, csv, APIs, databases each source have different structure and format which may make problems.

Question 6: What are APIs? Explain how APIs help in real-time data extraction.

Ans6: APIs Application programming interface which allows two different software to communicate with each other and transfer pr exchange data through it.

Help in real time data extraction:

- 1.Connect to data sources: APIs allow an ETL system to connect with applications and websites and databases or services.
- 2.Fetch real time data: uses updated data whenever it is requested so that data can be used for real time.
- 3.useful for live data: mainly used for live data such as stock price , weather , transactions and social media data.

Question 7: Why are databases preferred for enterprise-level data extraction?

Ans 7: Databases are preferred because they can store and handle large amounts of data efficiently, securely, and reliably.

Reasons :

1. Large Data Volume

Databases can handle millions or billions of records.

2. Fast Data Extraction

SQL queries can quickly retrieve only the required data.

3. Data Security

Databases provide authentication, permissions, and access control.

4. Data Consistency

They maintain accurate and consistent data using constraints and rules.

5. Easy Integration

Databases can easily connect with ETL tools, BI tools, APIs, and applications.

Question 8: What steps should an ETL developer take when extracting data from large CSV files (1GB+)?

Ans 8: When an ETL developer extracts data from a **large CSV file (1GB+)**, they should avoid loading the entire file into memory at once.

Steps:

1. Read the file in chunks

Divide the CSV into smaller parts and process them one by one.

2. Validate the data

Check for missing values, duplicate records, incorrect data types, and invalid rows.

3. Filter unnecessary data

Extract only the required columns and records to reduce processing time.

4. Handle errors

Identify and log corrupted or invalid records instead of stopping the entire process.

5. Transform the data

Clean and convert the required data into the correct format.

6. Load in batches

Insert the processed data into the target database in batches rather than all at once.

7. Monitor performance

Monitor memory usage, processing time, and errors during extraction.